

ASSIGNMENT 1

REQUIREMENT ANALYSIS

Submission Date: 07-September-2026

Course: B.Tech. CSE Semester: 5

Subject: System Design (TCS-504)

Project: Movie Ticket Booking System

A. Requirement Analysis

Functional Requirements

FR1: List Movies

The system shall display all movies currently playing in the cinema.

FR2: List Shows

The system shall display all available shows for a selected movie, including the screen number and start time.

FR3: Display Seat Availability

The system shall display the seat layout of a selected show and show whether each seat is **AVAILABLE** or **BOOKED**.

FR4: Book Seats

The system shall allow the customer to select one or more available seats for a show and shall reject the booking if any selected seat is already **BOOKED**.

FR5: Calculate Ticket Price

The system shall calculate the total booking amount according to the seat type: **SILVER = ₹150, GOLD = ₹250, and PLATINUM = ₹400**.

FR6: Make Payment

The system shall allow the customer to pay using **UPI, Card, or Cash**. If the payment fails, the booking shall not be confirmed and the selected seats shall be released.

FR7: Print Ticket

The system shall print a ticket containing **Booking ID, Movie name, Screen number, Show time, Seat numbers, and Total amount**.

FR8: Cancel Booking

The system shall allow a customer to cancel a confirmed booking, after which the booked seats shall become **AVAILABLE** again.

Non-Functional Requirements

NFR1: Modularity

The system shall be divided into separate classes, with each class responsible for one main task.

NFR2: Extensibility

The system shall allow a new payment method to be added without modifying the existing payment classes and booking flow.

NFR3: Input Validation

The system shall handle invalid menu choices, invalid seat numbers, already-booked seats, and invalid booking inputs with a clear error message without crashing.

NFR4: Maintainability

The code shall use meaningful names, avoid repeated code, and keep each function focused on one task.

This version keeps the **same structure and style as your submitted Part A**, while making the booking/payment failure requirement slightly more explicit and keeping the terminology consistent with the assignment.